

Production Node — Bill of Materials

24 GHz fixed-site drone detection and vehicle analytics node. Two-tier architecture: a 16-channel fan-beam array for long range at low elevation, plus a 4-channel wide-FOV module covering the overhead cone. Prototype column is quantity 5; volume is 10,000 units.

Specification

Parameter	Value
Band	24.075–24.175 GHz, FCC §15.245, unlicensed
EIRP	+32.7 dBm (regulatory ceiling)
Range, small quad (0.01 m2)	500 m crossing · 750 m hovering
Range, Phantom-class (0.03 m2)	660 m crossing
Range, overhead module	110 m in any direction
Coverage	90° az × 15° el at long range, uptilted 10° 120° × 120° overhead cone
360° coverage	4 nodes
Range resolution	1.5 m (100 MHz sweep)
Update rate	~1 Hz far, faster near
Ground targets	Vehicles and people on the same hardware
Power	PoE++ (802.3bt), < 25 W
Target price	\$2,000–2,500 per node + subscription

Why two tiers instead of one wide beam

Gain $\approx 26,000 / (\theta_{az} \times \theta_{el})$. A single $120^\circ \times 120^\circ$ beam is ~ 2.6 dBi — no gain, no range. Full 2D digital beamforming across that field of view needs 64 channels and still only reaches ~ 220 m. But a drone overhead is **close** — at 100 m altitude it is 100 m away, and R^4 gives back 28 dB versus 500 m. So: spend the aperture on the horizon where range is scarce, and cover the overhead cone with a cheap wide module. Same threat coverage, \$650 less at volume, and the 500 m low-elevation range is preserved.

Tier 1 — 16-channel fan-beam array

Long range, low elevation. Digital beamforming in azimuth, analogue subarrays in elevation.

Part	Qty	Function	Proto	Volume
ADF5904	4	4-channel 24 GHz RX downconverter, 10 dB NF	\$180	\$88
K-band LNA	16	≤2 dB NF, ≥20 dB gain — sets cascade NF	\$144	\$56
ADF5901	1	24 GHz TX MMIC, VCO, LO distribution to all RX	\$40	\$18
K-band PA	1	≥8 dB gain, P1dB ≥ +17 dBm — required , see note	\$25	\$10
ADF4159	1	Fractional-N PLL, linear FMCW ramp	\$18	\$8
TCXO ±0.5 ppm	1	Reference — phase noise caps integration gain	\$8	\$3
ADAR7251	4	16-bit 4-channel sigma-delta ADC	\$100	\$48
IF conditioning	16	Anti-alias filter + PGA per channel	\$10	\$4
RF passives, baluns, matching	—		\$25	\$6
Antenna board	1	8 vertical subarrays × 16 az channels + TX, Rogers RO4350B RF layer / FR4 stackup, 22 × 22 cm	\$380	\$70
Tier 1 subtotal			\$930	\$311

The PA is not optional

ADF5901 outputs **+8 dBm**. Into a 20 dBi TX array with 2 dB feed loss that radiates +26 dBm — **6.7 dB under the \$15.245 ceiling**, which is 1.5× range thrown away. An earlier revision of this BOM assumed +13 dBm from the chip and omitted the amplifier. Corrected here.

Tier 2 — 4-channel overhead module

120° × 120° wide field of view pointed up. Own illuminator, offset in frequency from Tier 1 so the two never interfere.

Part	Qty	Function	Proto	Volume
ADF5904	1	4-channel RX	\$45	\$22
K-band LNA	4	Same part as Tier 1	\$36	\$14
ADF5901	1	Second illuminator, offset frequency	\$40	\$18
ADAR7251	1	4-channel ADC	\$25	\$12
Antenna	1	2×2 wide-element patch array, ~9.5 dBi	\$95	\$42
IF conditioning, passives	—		\$19	\$7
Tier 2 subtotal			\$260	\$115

Compute, power, mechanical

20 channels total. Same silicon serves both tiers.

Part	Qty	Function	Proto	Volume
Zynq UltraScale+ ZU3EG	1	PL: LVDS deserialise, range/Doppler FFT, CFAR APU: tracking, fusion, network DPU: classifier inference	\$210	\$95
DDR4, 2 GB	1	Frame buffers — RAM is the constraint, not FLOPs	\$18	\$9
eMMC, 8 GB	1	Boot, models, local capture buffer	\$9	\$4
Gigabit PHY	1	Data out	\$5	\$2
Remote ID receiver	1	Cooperative-traffic filter + auto-labelling	\$50	\$18
Clock tree	—	20-channel distribution	\$22	\$8
Power	—	LDOs, switchers, sequencing, PoE++ PD	\$55	\$23
Main PCB	1	10-layer	\$95	\$11
Enclosure, radome, bracket, glands	—	IP66, polycarbonate radome, second window facing up	\$185	\$38
PCBA + 20-channel calibration	—	Per-unit assembly and cal time	\$290	\$52
Subtotal			\$939	\$260

Total

Block	Proto (qty 5)	Volume (10k)
Tier 1 — 16-channel fan beam	\$930	\$311
Tier 2 — 4-channel overhead	\$260	\$115
Compute, power, mechanical	\$939	\$260
Node total	\$2,129	\$686

Sale price	Gross margin at volume
\$2,000	66%
\$2,500	73%
\$3,000	77%

Four-node system

One node covers 90° azimuth. Four nodes give 360° at 500 m plus overlapping overhead coverage. At \$2,000 that is an **\$8,000 system** — against \$45,000–150,000 for three EchoGuard units, which is the incumbent configuration for 360°.

Costs not in the BOM

Item	Frequency	Cost
FCC equipment authorization	Once per design. A custom board voids the module grant.	\$25,000–40,000
Enclosure injection mould tooling	Once. Machine the first 100 instead.	\$10,000–30,000
Main board fab NRE	Per revision — budget 3 to 5	\$3,000–6,000
Antenna board NRE	Per revision. Rogers panel, larger board.	~\$3,500
EM simulation licence (HFSS / CST)	Free if UT ECE has it — confirm the IP position first	—
Test equipment — 24 GHz	VNA, signal analyser, near-field probe. Borrow from XRComm.	—

What a first spin actually costs

5 units at \$2,129 = \$10,645, plus roughly \$9,500 in NRE — about **\$20,000 per revision**. Three revisions is normal for a first RF design of this complexity, so budget **\$60,000 and 12–18 months** to a working production node. This is a post-funding activity. Prove the pipeline on ADI eval hardware first.

Do not cut

Damage here is irreversible — no amount of processing recovers it

Item	Why
LNA noise figure	Noise added at the front end is indistinguishable from signal forever. Removing the LNAs saves \$56 and costs 7 dB — 1.6× range.
TX power amplifier	6.7 dB of legally available EIRP left unused. 1.5× range.
Rogers on the RF layer	FR4 loss tangent about 0.02 versus 0.0037. Several dB of aperture efficiency, counted twice for two-way propagation.
TCXO phase noise	A cheap clock decorrelates the signal over long integration — it directly caps the 20 dB earned back in software.
Antenna element count	Aperture is physics. No processing substitute exists.

Cut freely

DSP genuinely covers these

Change	What it costs you	Saving (vol)
ADC 16-bit → 14-bit	Nothing. CFAR handles dynamic range better than extra bits.	~\$14
ZU3EG → Zynq-7020 + separate NPU	Tighter HDL, two chips instead of one	~\$25
Full Rogers → hybrid stackup	Nothing electrically	~\$18
Tooled enclosure → off-the-shelf IP66	Looks less finished	~\$15
Overhead module deleted	Loses the over-the-top case entirely. Do not — it is the cheapest coverage on the board.	~\$115

Verify before committing

#	Item
1	ADI chip pricing and lead time at 10k — negotiated, not listed
2	LNA selection: indie LNA_024_005 (3.2 dB NF, cheap) vs GaAs pHEMT (1.3–1.4 dB, costlier). Model the Friis cascade both ways against the range delta.
3	TX PA part selection — needs ≥ 8 dB gain at 24 GHz, P1dB $\geq +17$ dBm, and a package that fits the thermal budget
4	ADAR7251 maximum sample rate against the 4 MSPS requirement
5	Vitis AI DPU resource usage on ZU3EG alongside the FFT pipeline — both compete for the same DSP slices, and now for 20 channels
6	Rogers board pricing at panel quantity — the largest single line item at prototype scale
7	§15.245 measurement procedure for a swept emission — how the waveform is measured affects how much power you actually get